

0.1 Subsystem Engineering: Control and Actuation

0.1.1 Introduction

The Control and Actuation Subsystem serves as the intelligent decision-making core of an autonomous underwater vehicle deployed in the Antarctic Ocean. Its primary purpose is to control the vehicle's buoyancy by processing sensor data and commanding a motor-driven actuator to enable ascending, descending, or holding depth as required.

The engineering problem addressed is the need for a reliable, power-efficient, and fail-safe control system that can operate with minimal human intervention in extreme cold (-2°C to 0°C), under severe communication constraints, while guaranteeing payload safety and surface recovery at mission end or during critical failures. This section presents the requirement analysis, design choices, and verification of the Control and Actuation Subsystem in line with the overall project goals.

0.1.2 Requirements Analysis

The requirements for the Control and Actuation Subsystem were derived from the extreme operating conditions of the Antarctic Ocean, the safety-critical nature of the mission, and the need for seamless integration into existing AUVs that carry measurement platforms.

Key challenges addressed include near-freezing seawater temperatures, severely limited communication bandwidth, minimal real-time human intervention, and the fact that physical recovery after failure is often impossible. A core objective is the protection of the attached measurement platform, whose loss would mean the permanent loss of valuable oceanographic data.

User requirements were therefore prioritized around four key areas: reliability, payload protection and recovery, integration compatibility, and operational simplicity. These high-level requirements were systematically decomposed into detailed functional requirements, quantifiable design specifications, and verifiable acceptance test protocols. This structured flow-down ensures full traceability, technical feasibility, and alignment with the overall project goals within the constraints of a prototype development.

0.1.3 User Requirements

UR1	Reliability
Requirement	The subsystem must be highly reliable.
Rationale	In the remote Antarctic Ocean, system recovery after failure is often impossible. High reliability is therefore essential to protect the mission and the attached measurement platform.
Refined by	UR1:FR1 (High Uptime)
Verification	ATP1

UR2	Payload Protection and Recovery
Requirement	The subsystem must protect the attached measurement platform under all operating conditions and improve the likelihood of post-deployment AUV recovery.
Rationale	Loss or damage to the measurement platform would result in the permanent loss of irreplaceable oceanographic data. Reliable recovery of the AUV is therefore critical.
Refined by	UR2:FR1 (Autonomous Response); UR2:FR2 (End-of-Mission Surfacing); UR2:FR3 (Fail-safe Recovery)
Verification	ATP2; ATP3; ATP4

UR3	Integration Compatibility
Requirement	The subsystem must be compatible for integration with existing AUVs that carry measurement platforms.
Rationale	Existing AUVs were not designed with this subsystem in mind, meaning the electrical interface and communication protocols must conform to legacy hardware constraints rather than being designed from scratch.
Refined by	UR3:FR1 (Electrical Interface); UR3:FR2 (Communication Interface)
Verification	ATP5

UR4	Simple Operation
Requirement	The subsystem must be simple to configure and operate with minimal user interaction.
Rationale	In Antarctic field conditions, operators work under physical and time pressure with no opportunity to reconfigure the system once deployed. A misconfiguration at launch cannot be corrected remotely.
Refined by	UR4:FR1 (Single-step Activation); UR4:FR2 (Mission Time Monitoring)
Verification	ATP6; ATP7

0.1.4 Functional Requirements

UR1:FR1	High Uptime
Requirement	The subsystem shall maintain continuous firmware operation during deployment, provided that the supplied voltage remains within the specified operating range.
Refines	UR1 (Reliability)
Refined by	UR1:FR1:DS1 (Percentage Uptime)
Verification	ATP1

UR2:FR1	Autonomous Response
Requirement	The subsystem shall autonomously modify AUV manoeuvres (ascend, descend, or hold) based on input from its sensors.
Refines	UR2 (Payload Protection and Recovery)
Refined by	UR2:FR1:DS1 (Response Time)
Verification	ATP2

UR2:FR2	End-of-Mission Surfacing
Requirement	The subsystem shall ensure that the AUV ascends to the surface upon completion of the mission.
Refines	UR2 (Payload Protection and Recovery)
Refined by	UR2:FR2:DS1 (Mission Completion Reliability)
Verification	ATP3

UR2:FR3	Fail-safe Recovery
Requirement	The subsystem shall detect critical failure conditions (e.g., low battery) and command a recovery action to prevent the AUV and attached measurement platform from being stranded and lost.
Refines	UR2 (Payload Protection and Recovery)
Refined by	UR2:FR3:DS1 (Fail-safe Recovery)
Verification	ATP4

UR3:FR1	Electrical Interface
Requirement	The subsystem shall provide all connected hardware (sensing, power, and actuation) with a compatible electrical interface.
Refines	UR3 (Integration Compatibility)
Refined by	UR3:FR1:DS1 (Electrical Stability)
Verification	ATP5

UR3:FR2	Communication Interface
Requirement	The subsystem shall support communication with external sensors and payload electronics using defined protocols (UART and I2C).
Refines	UR3 (Integration Compatibility)
Refined by	UR3:FR2:DS1 (Communication Accuracy)
Verification	N/A

UR4:FR1	Single-step Activation
Requirement	The subsystem shall support single-step mission initiation via a user input.
Refines	UR4 (Simple Operation)
Refined by	UR4:FR1:DS1 (Activation Reliability)
Verification	ATP6

UR4:FR2	Mission Time Monitoring
Requirement	The subsystem shall monitor and maintain accurate tracking of elapsed mission time during operation.
Refines	UR4 (Simple Operation)
Refined by	UR4:FR2:DS1 (Mission Time Tracking Accuracy)
Verification	ATP7

0.1.5 Design Specifications

UR1:FR1:DS1	Percentage Uptime
Requirement	The subsystem shall achieve a system uptime of more than 99.5% during missions lasting up to 30 minutes, when operated in the specified environmental conditions.
Refines	UR1:FR1 (High Uptime)
Verification	ATP1

UR2:FR1:DS1	Response Time
Requirement	The subsystem shall command a change in AUV manoeuvres (ascend, descend, or hold) within 2 seconds of receiving sensor inputs from the actuator position sensor, battery level sensor and obstacle sensor.
Refines	UR2:FR1 (Autonomous Response)
Verification	ATP2

UR2:FR2:DS1	Mission Completion Reliability
Requirement	The subsystem shall achieve 100% platform surface recovery rate.
Refines	UR2:FR2 (End-of-Mission Surfacing)
Verification	ATP3

UR2:FR3:DS1	Fail-safe Recovery Behaviour
Requirement	At critical battery conditions (battery level $< 20\%$), the subsystem shall initiate ascent and maintain operation until the AUV has successfully surfaced.
Refines	UR2:FR3 (Fail-safe Recovery)
Verification	ATP4

UR3:FR1:DS1	Electrical Stability
Requirement	The subsystem shall maintain stable electrical operation ($3.3V - 5V$) without resets or overheating during continuous operation.
Refines	UR3:FR1 (Electrical Interface)
Verification	ATP5

UR3:FR2:DS1	Communication Accuracy
Requirement	The subsystem shall correctly receive and interpret all of the transmitted data from external sensors.
Refines	UR3:FR2 (Communication Interface)
Verification	N/A

UR4:FR1:DS1	Activation Reliability
Requirement	The subsystem shall correctly register a valid activation input with 100% success rate.
Refines	UR4:FR1 (Single-step Activation)
Verification	ATP6

UR4:FR2:DS1	Mission Time Tracking Accuracy
Requirement	The subsystem shall track elapsed mission time with an error not exceeding $\pm 5\%$ over the maximum mission duration of 30 minutes.
Refines	UR4:FR2 (Mission Time Monitoring)
Verification	ATP7

0.1.6 Acceptance Test Protocols

Code	Refines	Description
ATP1	UR1:FR1:DS1	The entire system is operated continuously for 60 minutes with software simulated sensor inputs. Pass if firmware uptime remains $\geq 99.5\%$ with no resets or failures.
ATP2	UR2:FR1:DS1	Simulated sensor inputs are applied to system and the delay between sensor input detection and actuator command is measured over multiple trials ($N = 10$). Pass if the response time is ≤ 2 seconds in all trials.
ATP3	UR2:FR2:DS1	A set of mock missions ($N = 10$) are executed to verify successful surface recovery upon mission completion. Pass if the AUV achieves surface recovery in 100% of trials under nominal conditions.
ATP4	UR2:FR3:DS1	Battery voltage levels above and below the critical threshold are simulated using a variable DC power source connected to the microcontroller's ADC. Pass if the AUV initiates ascent for all cases where battery level is $< 20\%$ of the supplied power and maintains operation until surfacing is complete.
ATP5	UR3:FR1:DS1	The subsystem is operated continuously for 60 minutes while the microcontroller ADC monitors its own temperature and the temperature of other subsystem components, and the microcontroller supply voltage is measured using a digital multimeter. Pass if voltage remains stable within $5V \pm 5\%$ with no abnormal overheating ($\Delta T < 10^\circ\text{C}$) or resets.
ATP6	UR4:FR1:DS1	Independent activation inputs (button presses, $N = 10$) are applied to an onboard push button on the microcontroller to verify correct mission initiation. Pass if 100% of valid inputs are correctly registered.
ATP7	UR4:FR2:DS1	The RTC elapsed time is measured by the microcontroller in software and is compared against an external digital stopwatch over a 30-minute interval. Pass if the timing error does not exceed ± 1.5 minutes ($\pm 5\%$).

Table 1: Acceptance Test Protocols for the Control and Actuation Subsystem

Refer to [Rationale for Parameter Values Used During Testing](#) in Appendix ?? for the justification and selection rationale of the values used during subsystem or system testing.

ATP1–ATP4 require full system integration, as they assess overall operational behavior arising from interactions between the sensing, power, and mechanical subsystems; successful verification therefore depends on coordinated system functionality. In contrast, ATP5–ATP7 were performed on the Control and Actuation Subsystem only, focusing on reliability, electrical stability, and timing accuracy. Isolating

these tests reduces external variability and enables a more accurate assessment of control hardware and firmware performance under controlled conditions.

0.1.7 Object Process Methodology

See figure 1 for an Object Process Methodology (OPM) diagram (i.e. a functional flow block diagram), representing the important components of the Control and Actuation Subsystem as well as how it interacts with other relevant subsystems mentioned in this report (namely, the power, sensing, and mechanical subsystems).

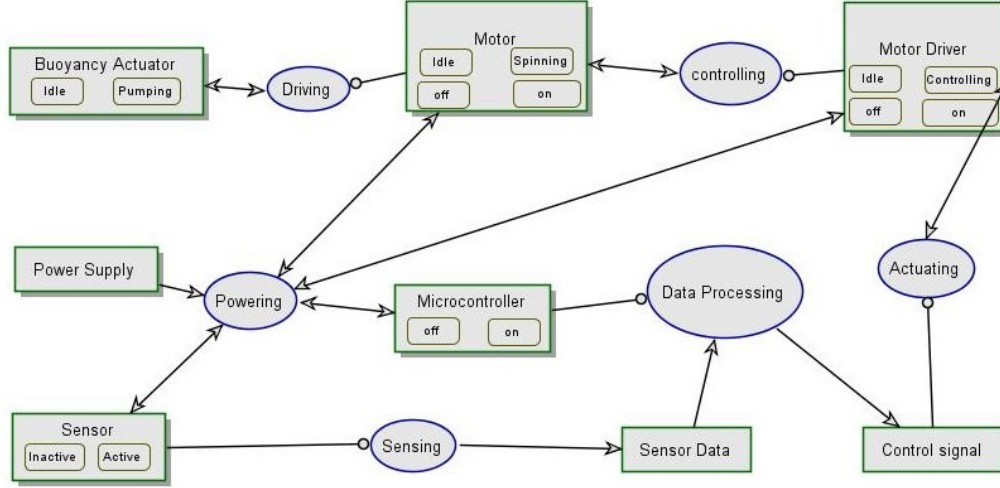


Figure 1: Object-Process Methodology (OPM) Diagram

See [Object-Process Methodology \(OPM\) Summary](#) in Appendix 0.2 for additional information on how the subsystems interact.

0.1.8 Atomic Level Solutions

Figure 2 shows the breakdown of the overall system into additional sub-components, until finally arriving at the 'atomic' level components.

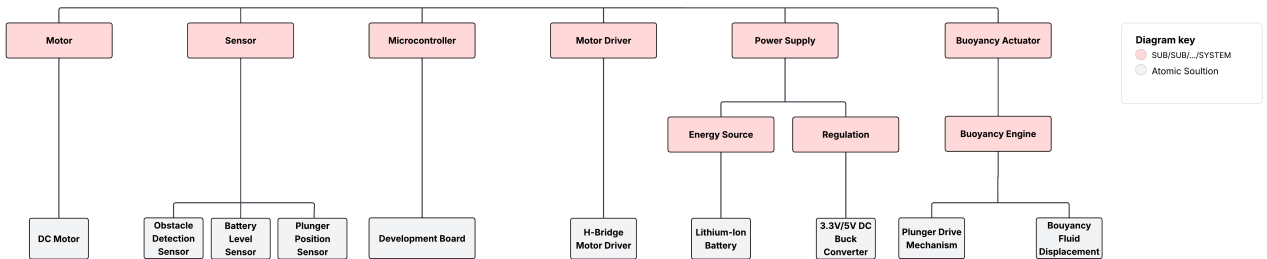


Figure 2: Atomic Level Breakdown for Control Subsystem

0.1.9 Atomic Level Analysis

Although numerous atomic components were identified throughout the overall architecture of the system, only the components associated with the control and actuation subsystem are analyzed in detail in the following section.

Development Board

The microcontroller forms a core element of the control and actuation subsystem, where it is responsible for data interpretation, decision-making, and generating actuator commands. In addition, it must accurately track real-world time to evaluate mission completion, which requires real-time clock (RTC) functionality.

Consequently, several candidate development boards were evaluated against key selection criteria, including processing performance, native hardware RTC support, and the availability of sufficient dedicated communication interfaces (I2C and UART/USART) for seamless integration with multiple peripheral sensors.

MCU	Clock	RTC	I2C	UART/USART
STM32F411 (Black Pill)	100 MHz (ARM Cortex-M4F)	Yes (LSE crystal and LSI)	3	3 USART
RP2040 (Pico)	133 MHz (dual ARM Cortex-M0+)	No (external needed)	2	2 UART
ESP32-S3	240 MHz (dual Xtensa LX7)	Software only	2	3 UART
SAMD51 (Feather M4)	120 MHz (ARM Cortex-M4F)	Yes	Up to 8 (via SER-COM)	Up to 8 (via SER-COM)
nRF52840	64 MHz (ARM Cortex-M4F)	Yes	2 TWIM	2 UARTE

Table 2: Comparison of Candidate MCUs

The STM32F411 was selected as it satisfied all three critical criteria: hardware RTC support (via both LSE and LSI sources), which also satisfies UR4:FR2; sufficient dedicated I2C and USART peripherals to serve all three sensors simultaneously without software multiplexing thus satisfying UR3:FR2; and deterministic Cortex-M4F core for reliable real-time actuator command generation. The RP2040 and ESP32-S3 were excluded primarily due to the absence of a hardware RTC, while the nRF52840 was excluded due to its comparatively limited clock speed.

DC Motor

For the selection of the motor to drive the buoyancy actuator, key features and specifications considered included self-locking capability, load torque, and stall current. The minimum torque required to displace seawater through the buoyancy mechanism was estimated analytically to verify actuator feasibility. The full derivation, based on external pressure, actuator geometry, and the actuator drive mechanism, is presented in Appendix 0.2. Table 3 presents a comparison of the leading motor options evaluated during the selection process.

Model	Load Torque	Load Current	Stall Current	Features
NEMA17	0.23 – 0.53Nm when energized and 0.012 – 0.025Nm when not powered	1.5A	1.5A	Not truly self-locking — can be back-driven
JGY370-10	1.37Nm	220mA	1.5A	Self-locking (worm gear prevents back-drive)
MG996R	0.89 – 1.08Nm	500 – 900mA	2.5A	No self-locking mechanism (holds via a PWM signal)

Table 3: Comparison of Candidate Motors

The JGY370-10 was selected as the actuator motor for its self-locking worm gear, which prevents back-driving and eliminates continuous holding power, and its high load torque relative to current draw. This torque was verified as sufficient using the analytical model in Equation 6 of the Appendix.

The NEMA17 stepper motor was rejected due to its lack of self-locking and high continuous current consumption, which would reduce energy efficiency. The MG996R servo was excluded because it requires continuous PWM holding torque, leading to higher power consumption and reduced positional security during power loss.

Motor Driver: H-Bridge Driver

Following motor selection, a suitable motor driver was required to ensure proper electrical interfacing and control. The primary selection criteria were supply voltage compatibility, output current capacity, operating temperature range, and power dissipation.

The JGY370-10 motor operates at a nominal supply voltage of 12V, with a typical running current of 220mA and a stall current of 1.5A. These values, particularly the stall current during high-load transients, were key drivers in the selection process. Power dissipation figures were obtained directly from manufacturer datasheets where available. For drivers specifying only $R_{DS(on)}$, dissipation was estimated using

$$P_D = I_{out}^2 \cdot R_{DS(on)} \quad (1)$$

under nominal load conditions. Table 4 compares the motor driver options evaluated.

Driver	Output Voltage	Output Current	Operating Temperature	Power Dissipation
L293D	0 – 36V	±600mA	0 – 70°C	5W
L298	0 – 50V	2A	–25 – 130°C	25W
TB6612FNG	0 – 15V	1.2A	–20 – 85°C	1.36W
DRV8871	–0.3 – 50V	3.5A	–40 – 125°C	13.6W
BTS7960	–0.3 – 45V	43A	–40 – 150°C	24mW

Table 4: Motor driver specifications and operating characteristics

Based on the operating environment (-2°C to 0°C), the L293D motor driver was removed as an option due to its thermal properties. The L298 was also excluded due to its relatively high power dissipation, which results in reduced efficiency under load. Ultimately, the BTS7960 was selected due to its high efficiency and superior current-handling capability, which makes it more suitable for low-temperature and energy-constrained operating conditions.

0.1.10 Embedded Firmware Architecture and Control Logic

The following section details the design of the firmware architecture and control logic implemented within the subsystem. The firmware is designed to prioritize determinism, reliability, and low resource usage.

Execution Model

The firmware operates using a bare-metal superloop architecture without an RTOS. This approach reduces software overhead, avoids concurrency-related issues, and ensures continuous evaluation of all sensors and control logic during system operation, supporting UR1. Within each loop iteration, the firmware performs actuator position acquisition, battery monitoring, obstacle detection, state machine evaluation, motor command generation, and RTC time update.

Control Logic

The control logic is implemented using a finite state machine (FSM) consisting of five states:

- IDLE: System is inactive and awaits mission activation input.
- SINKING: System commands descent until the target depth or condition is reached.
- HOLD: System maintains position with no motor actuation, relying on mechanical stability.
- SURFACING: System commands ascent until the surface condition is confirmed.
- FAILSAFE: System is triggered by critical conditions and forces safe recovery behavior.

The FSM enforces a deterministic mapping of all operating conditions to well-defined states, improving predictability and reliability. Each state represents a distinct operational mode of the system, with clearly defined entry conditions, behavior, and transition rules. Sensor inputs such as battery level, obstacle detection, and mission timing are continuously evaluated to determine whether a state transition is required. This structured approach prevents unclear control paths and ensures that, for any given set of inputs, the system behavior remains consistent and traceable.

0.1.11 ATP Results and Analysis

ATP1–Pass

ATP1 validated the continuous operational reliability of the AUV firmware under simulated sensor input conditions over a 60-minute run. The system achieved a firmware uptime of $\geq 99.5\%$ with no resets or failures recorded throughout the duration of the test, satisfying the pass criterion. This confirms that the firmware is stable under sustained operation and capable of handling continuous sensor input without degradation or unexpected interruptions.

ATP2–Fail

ATP2 assessed the real-time responsiveness of the system by measuring the delay between sensor input detection and actuator command across $N = 10$ trials. In all trials, the measured response time exceeded the 2-second pass threshold and therefore did not satisfy the system requirement. This was primarily attributed to communication latency introduced by the I2C and UART protocols, as delays between successive communication events were not sufficiently accounted for in the subsystem design. These protocol-related delays increased the overall response time of the subsystem.

Future iterations could improve performance through communication scheduling optimization, reduced blocking delays, or the implementation of interrupt-driven and non-blocking communication methods to minimize latency.

ATP3–Pass

ATP3 verified the AUV’s ability to reliably execute surface recovery at the conclusion of a mission. Across $N = 10$ mock mission trials conducted under nominal operating conditions, the AUV successfully surfaced in every trial, satisfying the pass criterion. The consistency of these results indicates that the recovery sequence is stable and repeatable, demonstrating dependable end-of-mission behavior under the tested conditions.

ATP4–Pass

ATP4 evaluated the subsystem’s emergency response to critically low battery conditions by applying voltages above and below the critical threshold using a variable DC power source connected to the microcontroller ADC. In every instance where the battery level dropped below 20% of the supplied voltage, the system correctly triggered an ascent procedure and sustained operation until surfacing was completed. These results validate the effectiveness of the low-power fail-safe mechanism and demonstrate that the system can maintain sufficient functionality during emergency recovery conditions.

ATP5–Pass

Design specification UR3:FR1:DS1 was successfully verified by ATP5. A digital multimeter was used to measure the microcontrollers voltage in time intervals of 10 minutes and temperature of the components of the subsystem was measured using a temperature sensor combined with the microcontrollers internal ADC. Table 5 below indicates the results recorded over the 6-10 minute intervals.

Time (s)	T_{measured} (°C)	T_{expected} (°C)	ΔT (°C)	V_{expected} (V)	V_{supply} (V)	ΔV (%)
10	21	16	5	5	5.11	2.20
20	19	16	3	5	5.11	2.20
30	20	16	4	5	5.10	2.00
40	21	16	5	5	5.10	2.00
50	19	16	3	5	5.10	2.00
60	19	16	3	5	5.10	2.00

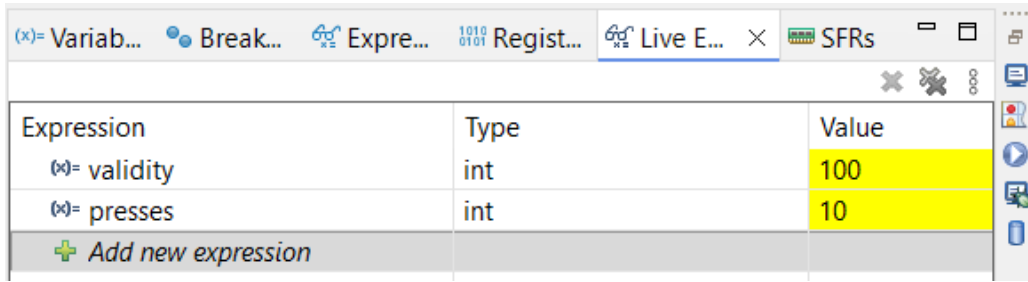
Table 5: ATP5 Test Results

The microcontroller’s voltage was confirmed to be around $5V \pm 5\%$ and the observed temperature of the subsystems components showed no signs of abnormal overheating over the test period thus

confirming electrical stability. Incorporating a limiter circuit into future iterations of the design may be beneficial, as certain components operate within strict voltage or current tolerances and therefore require protection against over-stress to prevent damage.

ATP6–Pass

ATP6 successfully verified requirement UR4:FR1:DS1 by demonstrating the reliability of the mission initiation interface under repeated user activation attempts. Independent activation signals were applied via a push-button interface, with software debounce filtering implemented to prevent duplicate triggers caused by mechanical contact. A valid activation was defined as the successful detection and registration of a single mission initiation event for each button press.



Expression	Type	Value
validity	int	100
presses	int	10
+ Add new expression		

Figure 3: Results for ATP7

The presses parameter represents the number of independent user activation inputs applied to the subsystem. The validity metric was determined from the proportion of correctly registered inputs over 10 independent activation attempts:

$$\text{Validity} = \frac{N_{\text{detected}}}{N_{\text{expected}}} \times 100$$

where $N_{\text{expected}} = 10$. All 10 activation attempts were correctly detected, resulting in a measured activation reliability of 100%.

ATP7–Pass

Both the RTC and the external stopwatch were triggered simultaneously at the start of the test. The stopwatch served as the ground truth reference to validate the actual time measured by the STM32F411’s RTC over the 30-minute mission duration.

RTC Measured Time (Minutes)	Stopwatch Time (Minutes)	Difference (Seconds)	Percentage Error (%)
02:00.00	02:05.08	5.08	4.23
04:00.00	04:10.87	10.87	4.53
10:00.00	10:28.25	28.25	4.71
16:00.00	16:44.85	44.85	4.67
24:00.00	25:07.06	67.06	4.66
30:00.00	31:24.31	84.31	4.68

Table 6: RTC timing accuracy test results

The RTC timing test showed a consistent systematic positive offset, with the internal clock running

slightly fast across all measured intervals. The percentage error remained stable between 4.2% and 4.7%, which is characteristic of the STM32F411's Low-Speed Internal (LSI) oscillator. All observed errors were well within the $\pm 5\%$ design tolerance specified in UR4:FR2:DS1. Consequently, ATP8 was successfully completed, verifying the mission time monitoring requirement.

For future iterations, it is recommended to replace the internal LSI oscillator with an external 32.768 kHz crystal oscillator to significantly improve long-term accuracy and reduce drift.

0.2 Appendix

0.2.1 Object-Process Methodology (OPM) Summary

The system operates as an integrated buoyancy control loop. The **Power Supply** powers the entire system via the **Powering** process, enabling the **Sensor** (which performs **Sensing** to generate **Sensor Data**), the **Microcontroller**, the **Motor Driver**, and the **Motor**. The **Microcontroller** executes **Data Processing** on the incoming Sensor Data to produce a **Control Signal**, an informational object that determines the behavior of the **Motor Driver**. The **Motor Driver**, when in its Controlling state, supplies controlled power to drive the **Motor** (which enters its Spinning state). The **Motor** then performs **Driving** to actuate the **Buoyancy Actuator**, switching it from Idle to Pumping as needed for buoyancy adjustment. Finally, **Actuating** translates the Control Signal into physical action within the system. This creates a closed-loop flow from sensing to actuation, all sustained by continuous powering.

0.2.2 Rationale for Parameter Values Used During Testing

ATP1 and ATP5 — 60-Minute Test Duration

The 60-minute test period was selected to represent a conservative upper bound on a single deployment mission, which is specified at 30 minutes maximum. By testing over a duration two times longer than the nominal mission, the protocol captures the effects of prolonged firmware execution, thermal build-up, and memory behavior that would not manifest in a short-duration test. A 99.5% uptime threshold over 60 minutes permits at most 0.3 minutes of cumulative downtime, which is sufficient to exclude transient, non-critical interruptions while still demanding near-continuous operation consistent with an autonomous ocean deployment. ATP5 applies the same duration criterion to electrical stability, ensuring that voltage regulation and thermal performance are evaluated under the same extended operating conditions.

ATP2 — 2-Second Response Time

The 2-second response time bound was chosen to ensure that the control system can respond to changes in sensor state faster than any meaningful physical change in platform buoyancy can occur. At typical actuation speeds, a 2-second delay is unlikely to result in an unsafe depth excursion. This threshold also aligns with human-perception timescales, ensuring that in any scenario requiring operator-observable confirmation, the system response appears instantaneous.

ATP3 — $n = 10$ Mock Missions, 100% Recovery

A sample size of $n = 10$ trials provides a statistically meaningful basis for evaluating mission completion reliability at the prototype stage, while remaining practically executable within laboratory conditions. A pass criterion of 100% reflects the safety-critical nature of surface recovery. The criterion is intentionally strict to provide confidence that the system behavior is repeatable and not dependent on favorable initial conditions.

ATP4 — Battery Threshold of <20%

The 20% battery threshold was selected to provide a sufficient energy reserve to complete a full ascent from maximum operating depth under worst-case conditions. Initiating recovery at 20% remaining capacity ensures that the system retains enough power to actuate the buoyancy mechanism, maintain firmware execution, and complete surfacing before the battery is fully depleted. This margin accounts for variability in battery discharge rates at low temperatures, consistent with the Antarctic operating environment.

ATP6 — $\geq 99\%$ Reception Accuracy

A 99% packet reception threshold reflects the minimum communication reliability required for the control system to maintain accurate sensor state awareness. Given that two of the three integrated sensors communicate via I2C and one via UART, a sustained packet error rate above 1% risks accumulating missed or corrupted sensor readings, which could lead to incorrect actuator commands. The 99% threshold therefore represents the minimum acceptable reliability below which control loop integrity cannot be guaranteed.

ATP7 — $n = 10$ Button Presses, 100% Registration

A sample of $n = 10$ independent activation inputs was chosen as a practical minimum for detecting systematic input registration failures, such as debounce errors or interrupt misses. The 100% pass criterion is applied because activation is a single-step, safety-critical operation — a missed activation in the field means the mission does not begin, with no opportunity for retry.

ATP8 — ± 1.5 Minutes Over 30 Minutes ($\approx \pm 5\%$)

The ± 1.5 -minute tolerance was selected to reflect a realistic upper bound on RTC drift achievable with the STM32F411's internal LSI oscillator under variable temperature conditions. This corresponds to approximately $\pm 5\%$ of the 30-minute mission duration. While tighter tolerances are achievable with an external LSE crystal, the $\pm 5\%$ bound ensures that mission timing remains operationally meaningful.

0.2.3 Motor Torque Requirement Derivation

The external pressure acting on the actuator was modeled as the sum of atmospheric pressure and hydrostatic pressure:

$$P_{\text{total}} = P_{\text{atm}} + \rho gh \quad (2)$$

where P_{atm} is atmospheric pressure, ρ is the density of seawater, g is gravitational acceleration, and h is the operating depth.

The resulting force acting on the actuator is:

$$F = P_{\text{total}}A \quad (3)$$

where the actuator cross-sectional area A is given by

$$A = \frac{\pi d^2}{4} \quad (4)$$

with d representing the internal diameter of the actuator.

Since the actuator employs a rack-and-pinion mechanism, the motor torque required to generate the corresponding linear force is

$$T = Fr \quad (5)$$

where r is the pitch radius of the pinion gear. Substituting the expressions for pressure, area, and force yields the final torque equation:

$$T = (P_{\text{atm}} + \rho gh) \frac{\pi d^2 r}{4} \quad (6)$$

This expression was used to determine the minimum torque requirement of the buoyancy actuator under expected operating conditions.